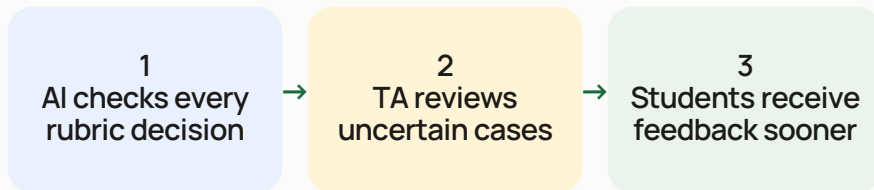


Problem: grading consumes the time meant for teaching

7 / 8
paid hours grading
measured TA
workday

AI makes the first pass; the TA teaches and resolves exceptions.



Sean Yépez + Dr. Mourad Bouache · Google

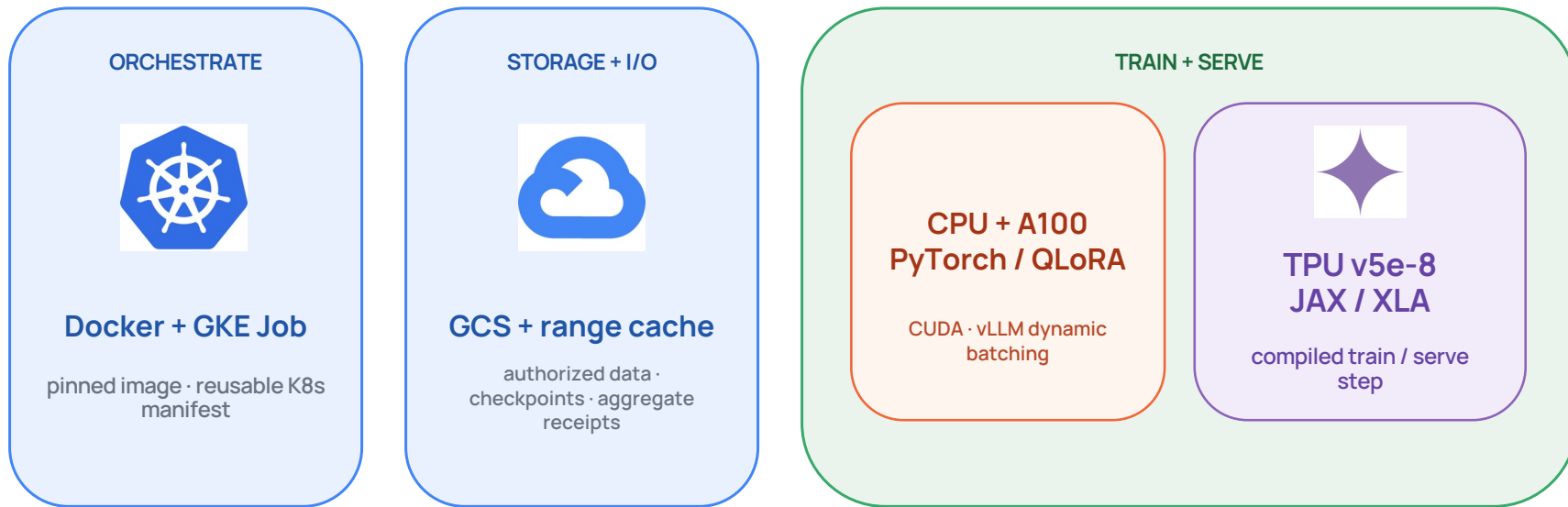
RiceChem supervised benchmark · 1,264 responses · 4 questions · 27 rubric items
8,392 TA-labeled TRUE/FALSE decisions become training, validation, and sealed test evidence.

RQ1 · QUALITY
Can tuning make a local model
accurate enough for rubric grading?

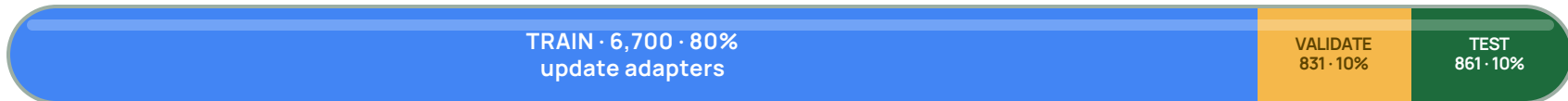
RQ2 · SYSTEMS
How do CPU, GPU, and TPU change
latency, throughput, and
utilization?

RQ3 · OPERATIONS
Which configuration meets the
quality bar and < 24 h at lowest
cost?

Containerize and run a supervised experiment on three compute lanes



EXPERIMENT DESIGN · 8,392 TA-LABELED DECISIONS



Validation selects the checkpoint; the test split stays sealed.

Measurements: instrument the workload—and disclose what is missing

CONTROLLED WORKLOAD

861 held-out rubric decisions

same manifest · prompt · parser · output cap · checkpoint

concurrency 1 + 24

EXECUTION LANES

RUNTIME

CAPTURED SIGNALS

CPU

16 vCPU baseline

Transformers

host memory · process telemetry

CPU utilization

peak RSS

p50 / p95

decisions / s

GPU

one A100 40 GB

PyTorch · CUDA · vLLM

NVML · batching · endpoint timing

GPU utilization

peak VRAM

load time

tokens + decisions / s

TPU

v5e-8 slice

JAX / XLA · vLLM-TPU

XProf · compile cache · GCS-FUSE

TPU utilization

peak HBM

compile time

cache + I/O

COMPARISON GATE

Compare hardware only when checkpoint, prompt, parser, output cap, batching, and concurrency match.

WORKLOAD

861 decisions · same prompt/parser/output cap · c=1 and c=24

COMPUTE

wall time · p50/p95 · decisions/s · wall/step time as cycle proxy

MEMORY

CPU mCPU + RSS · A100 NVML + VRAM · TPU host memory only

INPUT / I/O

GCS range cache fixed startup; load and XLA compile time not isolated

TPU device utilization + HBM were not enabled in this run. Remedy: add XProf / TensorBoard TPU profiling (or Cloud TPU device metrics), profiler hooks, and service-account permissions; report unavailable values rather than estimate them.

Accelerators cut single-stream grading time 30–41×

THROUGHPUT SPEEDUP · CONCURRENCY 1 · CPU = 1×

16-vCPU



0.28/s · 50.9 min · 95.1% CPU · 18.0 GB RSS

A100 40 GB



8.54/s · 100.8 s · 19.5% GPU · 35.6 GB VRAM

TPU v5e-8



11.39/s · 75.6 s · device util/HBM unavailable · 54 GB host

CONCURRENT SERVING · c = 24

A100 123.9/s

\$0.007 per 861 decisions

TPU 98.4/s

\$0.023 per 861 decisions

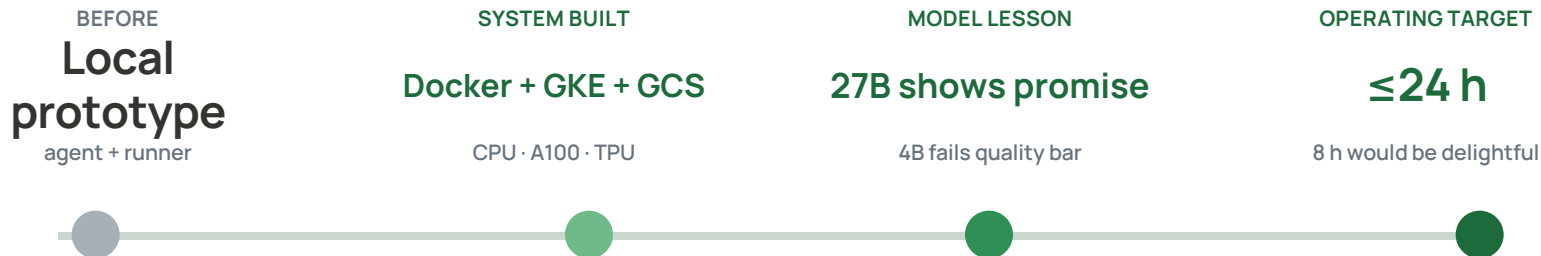
These are one-endpoint measurements. Add GKE runners/endpoints as queue depth grows; cluster throughput is bounded by quota and coordination overhead.

QUALITY GATE · ~72% AGREEMENT

Gemma 4B failed the practical grading bar.

TA re-review overhead would erase the throughput benefit.

ME344 moved Treemarks from a local prototype to a repeatable cloud ML experiment



Meet the grading-quality threshold → choose the smallest system
under 24 hours → minimize cost.

BOTTLENECK

27B serving was unbatched; training also exposed accelerator-memory and object-store startup pressure.

NEXT · batch the 27B endpoint; profile 27B on TPU; test leave-one-question-out transfer; compare per-assignment training economics with frontier inference; add runners only when queue depth threatens the 24-hour promise.